Grounding Lifelong Learning Dialogue Through Retrieval-Augmented Generation and Hallucination Stress Testing

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***Abstract*—Domain-specific chatbots can provide useful information from organizational documents, but unrestricted language generation may produce responses that are unsupported by the available knowledge base. This study implemented and evaluated a Retrieval-Augmented Generation (RAG) chatbot for the lifelong learning domain. The system loads two PDF documents, divides them into 500-character chunks with 50-character overlap, generates embeddings using all-MiniLM-L6-v2, stores them in ChromaDB, retrieves the three most relevant chunks, and generates responses through OpenAI GPT-OSS 120B accessed using Groq. The baseline configuration used temperature 0 and an explicit fallback rule. An out-of-domain query correctly produced the required fallback response. During the stress test, temperature was increased to 1.0 and the fallback instruction was removed; the model still refused to provide unsupported information. The results demonstrate the value of retrieval grounding and prompt restrictions in reducing unsupported chatbot responses.**

Keywords—retrieval-augmented generation, lifelong learning, hallucination, embeddings, ChromaDB, GPT-OSS 120B, LangChain

I. SYSTEM ARCHITECTURE

The implemented system is a domain-specific Retrieval-Augmented Generation (RAG) chatbot designed to answer questions using a knowledge base focused on lifelong learning. RAG combines retrieval from an external knowledge source with language generation, allowing the model to condition its response on retrieved information rather than relying only on knowledge stored in its parameters [1]. The implemented pipeline consists of document loading, text chunking, embedding generation, vector storage and retrieval, prompt-based context injection, and language-model generation.

The selected domain is Lifelong Learning, aligned with topics related to adult learning, continuing education, participation, inclusion, skills development, and Sustainable Development Goal 4. The knowledge base contains two PDF files: 381666eng.pdf and UIL_Handbook_170x240_RZ220601_Online.pdf. The notebook reports that these two documents were successfully loaded and divided into 2,491 chunks.

A. Document Loading

The first stage uses LangChain's DirectoryLoader to load files from the my_data/ directory. The loader is configured with the pattern **/*.*, allowing files within the directory and its subdirectories to be processed. The notebook successfully loaded two raw documents.

The use of an external document collection is important because the chatbot is intended to answer questions about a specific domain rather than act as a general-purpose question-answering system. The uploaded documents therefore serve as the primary knowledge source for the chatbot.

B. Document Chunking

After loading, the documents are divided using RecursiveCharacterTextSplitter. The implemented configuration specifies a chunk size of 500 characters and an overlap of 50 characters.

Chunking converts long documents into smaller searchable units. The overlap preserves a portion of the preceding chunk so that information occurring near chunk boundaries is less likely to lose contextual continuity. The notebook produced 2,491 chunks from the two source documents.

C. Embedding Generation

Each document chunk is transformed into a numerical representation using the **all-MiniLM-L6-v2** embedding model through HuggingFaceEmbeddings. Sentence-transformer models are designed to represent text in an embedding space where semantic similarity can be used for retrieval [2].

The selected MiniLM model provides a relatively lightweight embedding approach suitable for semantic retrieval. The notebook uses the model directly without additional fine-tuning.

D. ChromaDB Vector Storage and Retrieval

The generated embeddings are stored in **ChromaDB** using Chroma.from_documents(). The vector database provides the searchable representation of the document collection.

The notebook then converts the vector store into a retriever using:

```
search_kwargs={"k": 3}
```

Therefore, the retriever returns the **three most relevant chunks** for a user query. These retrieved chunks become the context supplied to the language model.

This retrieval stage is central to the RAG design because the model receives information selected from the uploaded knowledge base before generating an answer. The approach follows the general principle of combining a generative model with external non-parametric memory [1].

E. Prompt and Language Model

The original chatbot uses ChatGroq with the model identifier **openai/gpt-oss-120b** and a temperature of 0. The system prompt explicitly instructs the model to answer only from the provided context:

"Answer questions strictly using ONLY the provided context below."

The prompt also contains the fallback rule:

"If the answer cannot be found in the context, reply: 'I cannot answer based on the provided domain data.'"

The retrieved chunks are inserted into the {context} placeholder. The user's question is supplied through the {input} variable.

The notebook therefore implements a constrained generation process in which retrieval provides the evidence and the system prompt restricts the model from relying on unsupported information.

F. RAG Chain Assembly

The pipeline is assembled using create_stuff_documents_chain() and create_retrieval_chain(). The document-combining chain connects the retrieved context, prompt, and language model. The retrieval chain then connects the retriever to the document-processing chain.

The resulting flow is:

Documents → Chunking → Embeddings → ChromaDB → Top-3 Retrieval → Context Prompt → GPT-OSS 120B → Response

This structure represents the implemented RAG pipeline rather than an assumed architecture.

II. IMPLEMENTATION DETAILS

TABLE I. IMPLEMENTATION PARAMETERS

Parameter	Implemented Configuration
Domain	Lifelong Learning
Source documents	381666eng.pdf; UIL_Handbook_170x240_RZ220601_Online.pdf
Raw documents loaded	2
Generated chunks	2,491
Chunk size	500 characters
Chunk overlap	50 characters
Embedding model	all-MiniLM-L6-v2 via HuggingFaceEmbeddings
Vector database	ChromaDB
Retrieval value	Top 3 (k=3)
Baseline temperature	0
Baseline fallback	Enabled
Baseline grounding instruction	Answer only using provided context
Stress-test temperature	1.0
Stress-test fallback	Removed
Stress-test grounding instruction	Context-only instruction retained
Baseline LLM	openai/gpt-oss-120b via Groq
Stress-test LLM	openai/gpt-oss-120b via Groq
API-key handling	getpass and GROQ_API_KEY environment variable

The implementation also includes secure API-key handling. Instead of embedding the Groq API key directly in the source code, the notebook checks whether GROQ_API_KEY exists in the environment and, if necessary, requests it through getpass. This prevents the key from being directly displayed in the notebook code.

The notebook also explicitly installs the required LangChain, Groq, Hugging Face, ChromaDB, PDF-processing, and document-processing packages. The implementation does not contain an additional reranker, keyword-search layer, manually coded similarity calculation, or separate hallucination classifier; therefore, these components are not claimed as part of the system.

III. EXPERIMENTAL RESULTS AND HALLUCINATION ANALYSIS

The experiment evaluated the chatbot using an in-domain query and an out-of-domain query. The required hallucination stress test used the same out-of-domain question under two configurations.

A. In-Domain Test

The first test used:

Query: "Summarize the key information found in the documents."

The retrieved sources were:

1. my_data/381666eng.pdf
2. my_data/UIL_Handbook_170x240_RZ220601_Online.pdf
3. my_data/381666eng.pdf

The generated response discussed a "Structure of this report," an "lll initiative," copy-editing by Alec McAulay, and design by Christiane Marwecki.

The output demonstrates that retrieval was functioning because the system returned information originating from the retrieved documents. However, the response was not well aligned with the broad request to summarize the key information about lifelong learning. This indicates that successful retrieval does not automatically guarantee that the generated answer will be optimally relevant to the user's intent. The result should therefore be interpreted as a limitation of the tested retrieval-generation configuration rather than as evidence of complete domain-answering accuracy.

B. Out-of-Domain Baseline Test

The out-of-domain query was:

"What are the planets in the solar system and how do they orbit the sun?"

The expected behavior was to refuse to answer because information about the solar system was not part of the lifelong learning knowledge base.

Under the baseline configuration, which used **temperature = 0** and the explicit fallback instruction, the actual output was:

"I cannot answer based on the provided domain data."

This result directly followed the fallback rule specified in the system prompt. No unsupported information about planets or orbital motion was generated.

C. Hallucination Stress Test

For the stress test, the same query was executed after changing the configuration.

The temperature was changed from **0 to 1.0**, and the explicit fallback instruction was removed. The context-only instruction remained in the system prompt.

The actual response became:

"I'm sorry, but the information you've provided doesn't include any details about the planets in the solar system or how they orbit the Sun."

The wording differed from the baseline response, but the model still refused to provide unsupported information.

D. Experimental Comparison

TABLE II. RECORDED TEST QUERIES, EXPECTED RESPONSES, AND ACTUAL MODEL OUTPUTS

Test	Query	Expected Response	Actual Model Output
In-domain	"Summarize the key information found in the documents."	A relevant summary of the lifelong learning documents	Generated a summary concerning report structure, an "lll initiative," copy-editing, and design credits
Baseline out-of-domain	"What are the planets in the solar system and how do they orbit the sun?"	Refuse using the specified fallback message	"I cannot answer based on the provided domain data."
Stress test	Same solar-system query	Avoid unsupported information despite increased temperature and removed fallback	"I'm sorry, but the information you've provided doesn't include any details about the planets in the solar system or how they orbit the Sun."

TABLE III. BASELINE AND STRESS-TEST COMPARISON

Evaluation Aspect	Baseline	Stress-Test Configuration
Temperature	0	1.0
Explicit fallback rule	Enabled	Disabled
Context-only instruction	Enabled	Enabled
Query	Same out-of-domain query	Same out-of-domain query
Response behavior	Explicit fallback	Context-based refusal
Unsupported solar-system facts	None observed	None observed
Wording consistency	Exact predefined fallback	Different generated refusal
Hallucination in tested query	Not observed	Not observed

The comparison shows an important distinction between **hallucination resistance** and **response control**. The baseline configuration produced the exact predefined fallback message. This made the system behavior predictable when information was missing.

The stress-test configuration was less restrictive because the explicit fallback instruction was removed and the temperature was increased to 1.0. Nevertheless, the model still recognized that the retrieved context did not contain information about the solar system and declined to answer. Thus, **no hallucination was observed in this particular stress-test execution**.

The experiment does not justify the broader conclusion that temperature 1.0 prevents hallucination. It only demonstrates that, for the tested query and retrieved context, the model continued to refuse an unsupported request. A larger evaluation set would be required to measure hallucination frequency statistically.

The in-domain result also provides an important observation. Although the system successfully retrieved document chunks, the generated response was only partially relevant to the requested summary. This demonstrates that grounding is not equivalent to answer quality. A response may contain information originating from retrieved documents while still failing to address the user's intended question.

IV. CONCLUSION

The implementation and testing of the Lifelong Learning RAG chatbot demonstrated how retrieval, prompt constraints, fallback behavior, and secure configuration practices can work together to produce more controlled chatbot behavior.

First, retrieval-based information provides an external knowledge source that limits the information supplied to the language model. Instead of depending exclusively on the model's internal knowledge, the system retrieves relevant chunks from the uploaded documents before generating an answer. This reflects the central motivation of RAG systems, which combine language generation with access to external non-parametric memory [1].

Second, prompt restrictions provide an additional layer of control. The instruction to answer strictly from the provided context establishes a clear boundary for the model. The baseline fallback rule strengthens this boundary by

explicitly defining what should happen when the retrieved context does not contain the requested information.

Third, the experiment demonstrated that fallback rules improve predictability. With temperature 0 and the fallback rule enabled, the chatbot returned the exact expected fallback response for the solar-system query. When the fallback rule was removed and temperature increased to 1.0, the response changed in wording, but the model still avoided unsupported information in the tested case.

Fourth, the experiment showed that retrieval grounding alone does not guarantee perfect relevance. The in-domain summary test retrieved the correct source files but produced a response focused on particular document details rather than providing a comprehensive summary of the lifelong learning content. This indicates that future evaluation should consider not only whether a response is grounded but also whether it directly answers the user's question.

Finally, secure API-key practices are necessary when connecting applications to external model providers. The notebook uses getpass and an environment variable rather than placing the API key directly in the source code. This reduces the risk of accidentally exposing credentials in shared notebooks.

Overall, the practical implementation showed that safer and more reliable RAG systems require multiple complementary controls. Retrieval supplies domain evidence, prompt restrictions establish behavioral boundaries, fallback rules handle missing information, and secure credential handling protects the external service connection. The results further demonstrate that hallucination evaluation should be based on actual experimental observations rather than assumptions about model behavior.

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